EXERCISES WEEK 7

8CM10 - Introduction to Quantum Physics and Structure of Matter

NUCLEAR MASSES AND BINDING ENERGIES

1. Compute the binding energy B_{tot} and the binding energy per nucleon B_{tot}/A for:

   a) $^{208}\mathrm{Pb}$
   b) $^{133}\mathrm{Cs}$
   c) $^{90}\mathrm{Zr}$
   d) $^{59}\mathrm{Co}$

   For the atomic masses, check the internet or the tables at the end in the book of Krane.

RADIOACTIVE DECAY

2. A certain sample of a radioactive material decays at a rate of 548 per second at $t = 0$. At $t = 48$ minutes, the counting rate has fallen to 213 per second.
   a) What is the half-life of the radioactivity?
   b) What is its decay constant?
   c) What will be the decay rate at $t = 125$ minutes?

3. Ordinary potassium contains 0.012 percent of the naturally occurring radioactive isotope $^{40}\mathrm{K}$, which has a half-life of $1.3 \cdot 10^9$ y.
   a) What is the activity of 1.0 kg of potassium?
   b) What would have been the fraction of $^{40}\mathrm{K}$ in natural potassium $4.5 \cdot 10^9$ y ago?

DECAY PROCESSES

4. The natural decay chain $^{238}_{92}\mathrm{U} \rightarrow {}^{206}_{82}\mathrm{Pb}$ consists of several alpha decays and negative beta decays.
   a) How many alpha decays must occur in the chain?
   b) For that number of alpha decays, how many beta decays must occur to make the final Z come out correctly?
   c) Sometimes a decay chain can proceed through different branches and thus have different intermediate radionuclides. Do your answers to a) and b) depend on this branching?

5. Using the table below, show that $^{8}_{4}\mathrm{Be}$ can decay to two α-particles, but $^{12}_{6}\mathrm{C}$ cannot decay to three α-particles. How much energy is released at decay?

Nucleus	Binding Energy (MeV)	Binding energy per nucleon (MeV/A)
$^{2}_{1}\mathrm{H}$	2.22	1.1
$^{4}_{2}\mathrm{He}$	28.30	7.1
$^{8}_{4}\mathrm{Be}$	56.50	7.1
$^{12}_{6}\mathrm{C}$	92.16	7.7

6. Complete the following decays and compute the corresponding Q-values:

   a) $^{27}\mathrm{Si} \rightarrow {}^{27}\mathrm{Al} + \cdots$
   b) $^{93}\mathrm{Mo} + e^{-} \rightarrow \cdots$

7. The atomic mass of $^{211}_{84}\mathrm{Po}$ is 210.986627 u, and it decays by α-emission to $^{208}_{82}\mathrm{Pb}$, which has an atomic mass of 206.975872 u.
   a) Calculate the decay energy Q and compare this to the rest energy of the least massive particle. Can we reasonably conclude that we may ignore relativistic effects in this fission reaction? *(Note: Your answer here holds in general for α-emissions, even though you only analyze it in this case)*
   b) Assuming that the parent has no kinetic energy before the reaction and that all of the decay energy is distributed between the α-particle and daughter nucleus, derive the formula that gives the distribution of Q between α-particle and daughter.
   c) Calculate the actual kinetic energies of the α-particle and daughter nucleus. Compare the kinetic energy of the α-particle with the value on the chart of nuclei.

8. Construct a decay diagram for $^{224}_{90}\mathrm{Th}$, which decays by α-emission to $^{220}_{88}\mathrm{Ra}$. The decay energy Q is 7.305 MeV, the half life is 1.045 s and the energies of emissions are given as:

Emission	Energy (MeV)	Abundance (%)
α_1	7.170	79
α_2	6.997	19
α_3	6.768	1.2
α_4	6.702	0.6
γ_1	0.177	
γ_2	0.235	
γ_3	0.297	
γ_4	0.410	

9. The fission of uranium-235 can occur in several ways. One possible fission reaction is $^{235}\mathrm{U} + n \rightarrow {}^{141}\mathrm{Ba} + {}^{92}\mathrm{Kr} + 3n$ where n is a neutron. The reaction produces 3 neutrons which may give rise to new fission reactions.
    a) Compute the energy that is produced ("Q value") by one fission reaction of this type.
    b) Compute the total amount of energy in Joules if all atoms in 235 gram of $^{235}\mathrm{U}$ perform this reaction.
    c) The heat of combustion of methane ($\mathrm{CH_4}$) is about 889 kJ/mol. How much energy (in joules) is produced if 235 grams of methane are combusted?